

Q1: NOV 2002 P2

- 2) A basketball player shoots a 'free throw' as shown in Fig 2.1 below. He throws the ball at an angle of 55° to the horizontal and gives it an initial velocity u .

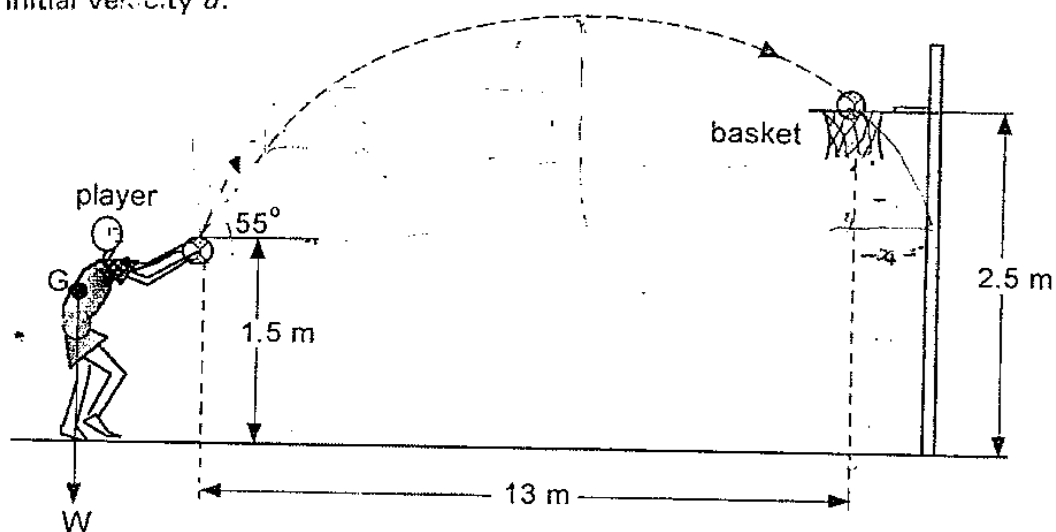


Fig. 2.1

Assume that air resistance is negligible.

- (a) What is the component of the ball's initial speed in terms of u ?
- (i) horizontally _____
- (ii) vertically _____ [2]
- (b) Using your answer to a(i), write down an expression for the time t , taken to reach the basket in terms of u .

$t =$ _____ [1]

- (c) Using your answers to (a) and (b), calculate the value of u .

Q2: JUN 2003 P2

- 2 (a) (i) Define displacement.

.....

.....

- (ii) Use your definition to explain how it is possible for a car to travel a certain distance and yet have zero displacement.

.....

.....

[3]

- (b) A car starts from rest and travels upwards along a straight road inclined at an angle of 5.0° to the horizontal, as illustrated in Fig. 2.1.

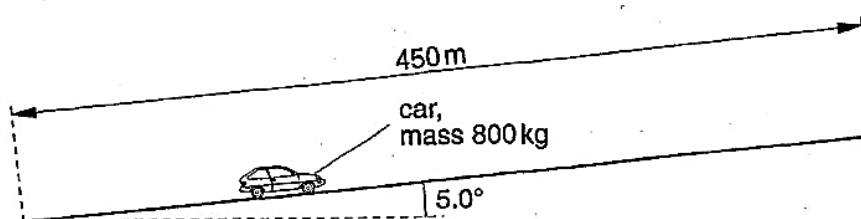


Fig. 2.1

The length of the road is 450 m and the car has mass 800 kg. The speed of the car increases at a constant rate and is 28 m s^{-1} at the top of the slope.

- (i) Determine, for this car travelling up the slope,

1. its acceleration,

acceleration = m s^{-2} [2]

2. the time taken to travel the length of the slope,

Q3:JUN 2008 P2

- 1 Fig. 1.1 shows a tank being used to collect water from the roof top. The down pipe was bent so that its open end is 0.8 m away from the wall.

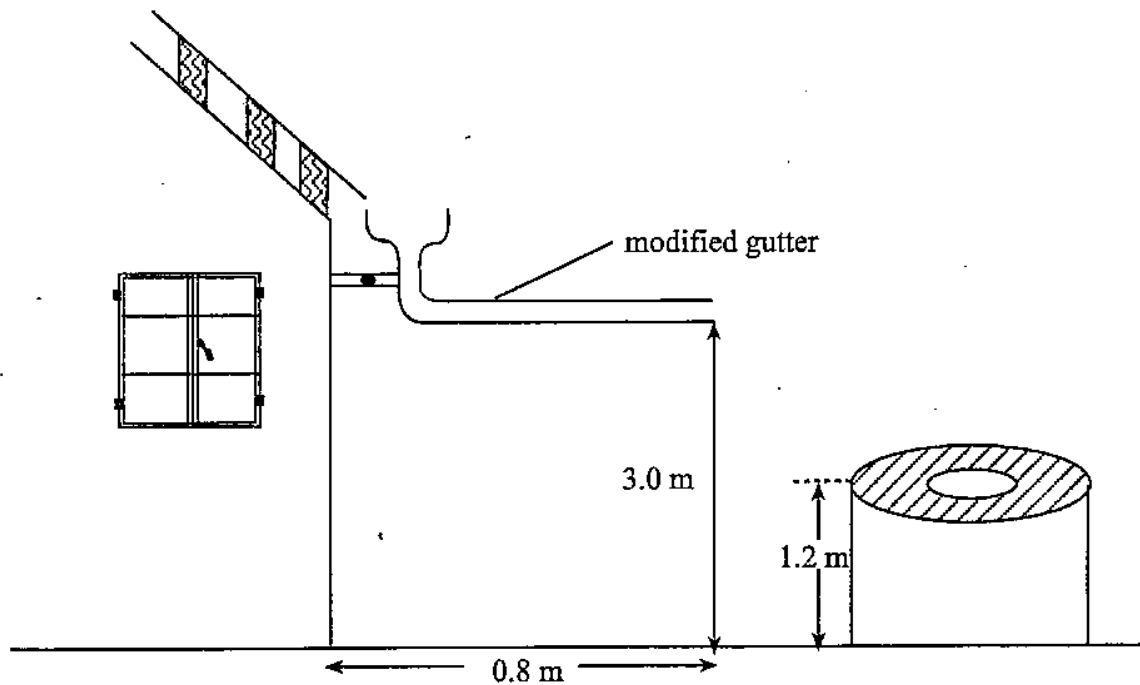


Fig. 1.1

- (i) Show on Fig. 1.1 the path followed by water gushing out from the central part of the modified gutter onto the central opening on the tank. [1]
- (ii) Given that the water leaves the gutter with a velocity of 2.0 ms^{-1} , determine
 1. the time it takes for the water to reach the mouth of the tank,

2.1 KINEMATICS P2&P3 QUESTIONS

Compiled by: R. Nkomo

2. The distance of the centre of the tank from the wall.

distance = _____ [3]

Q4: NOV 2012 P2

- 2 (a) Define

(i) *distance*,

(ii) *displacement*,

(iii) *acceleration*.

[3]

- (b) Some of Newton's equations of motion are as follows:

$$v = u + at$$

$$s = \frac{1}{2}(u + v)t$$

$$v^2 = u^2 + 2as$$

2.1 KINEMATICS P2&P3 QUESTIONS

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State the **two** conditions which are considered in the derivation of the equations.

[2]

- (c) Explain how it is possible for a car to move a distance of 50 km but with zero displacement.

[1]

Q5:NOV 2014 P2

- 1 A motorist is moving along a 2 m wide road at 40 ms^{-1} . He spots a wild pig 5 m away from the road moving at 2.5 ms^{-1} .

If their direction of travel are at right angles, calculate the time taken by the pig

- (i) to reach the road,

time = _____

- (ii) to cross the road.

2.1 KINEMATICS P2&P3 QUESTIONS

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- (iii) The motorist applied brakes after 0.5 s some 10 m away from the pig's crossing point to avoid hitting it. The mass of the motorist and his car is 12 000 kg.

State the motorist's reaction time.

- (iv) Determine the distance from where the motorist noticed the pig to the crossing point.

distance = _____

[6]

Q6: NOV 2015 P2

- 2 (a) Define *acceleration*.

[1]

- (b) A ball of mass 50.0 g is thrown vertically downwards with a velocity of 3.25 ms^{-1} . It hits the ground after 1.62 s.

- (i) Assuming air resistance to be negligible, calculate the

1. speed of the ball when it hits the ground,

speed _____

2. distance travelled by the ball to the ground.

Q7:JUN 2018 P2

- 1 (a) Define the term *displacement*.

[1]

- (b) An aeroplane is travelling at a steady speed of 40 ms^{-1} at an altitude of 250 m. A metal sphere is released from the aeroplane directly above a point on the level ground.

[Neglect air resistance]

Calculate the

- (i) time of flight of the sphere,

time of flight = _____

- (ii) speed on impact of the sphere,

speed on impact = _____

- (iii) distance from the point the sphere is released to the point of impact.

distance = _____ [6]

Q8:NOV 2003 P3

- 1 (a) Distinguish between distance and displacement. [3]
- (b) A ball is projected at an angle θ to the horizontal from a cliff 96 m high and strikes the ground at a distance of 260 m away from the cliff as shown in Fig. 1.1. The time of its flight is 7.0 s.

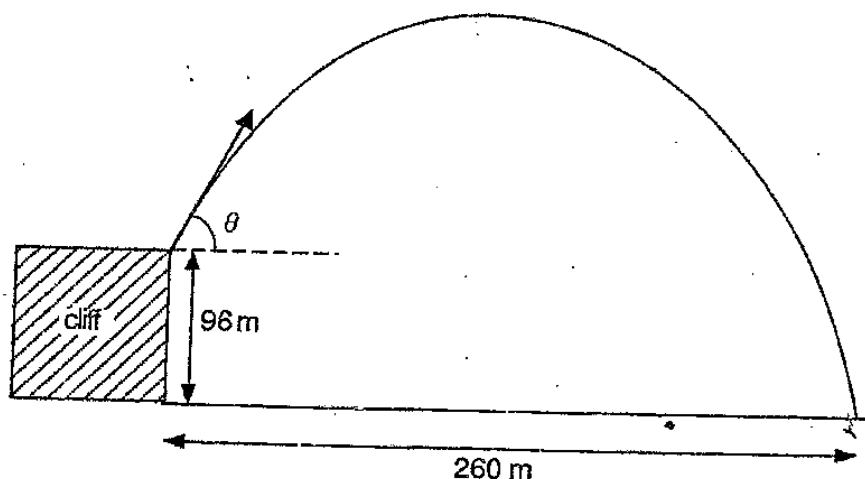


Fig. 1.1

If it is assumed that air resistance is negligible, calculate

- (i) the initial vertical velocity of the ball,
 - (ii) the time to reach the greatest height,
 - (iii) the initial horizontal velocity,
 - (iv) the angle θ ,
 - (v) the velocity of the projectile on hitting the ground.
- [11]
- (c) (i) On the same axes, sketch graphs to show the variation with time of
- 1 the horizontal velocity,
 - 2 the vertical velocity of the ball in (b).
- [3]
- (ii) Explain how the graphs may illustrate
- 1 the vertical distance travelled by the ball,
 - 2 the vertical displacement of the ball,
 - 3 the horizontal displacement of the ball.
- [3]

2.1 KINEMATICS P2&P3 QUESTIONS

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Q9: NOV 2011 P3

- 3 (a) Define *deceleration* and *velocity*. [2]
- (b) An underground train moved between two stations. It started from rest and accelerated uniformly for 51 seconds until it reached 20 ms^{-1} . It then maintained this velocity for the next 78 seconds and then decelerated uniformly to rest for 21 seconds.
- Sketch a
- (i) velocity-time graph,
- (ii) distance-time graph for the train. [4]
- (c) Objects falling towards the Earth's surface under gravity are said to be in free fall.
- State and give a reason whether the following are in free fall.
- (i) A parachutist falling with the parachute open.
- (ii) A steel ball bearing falling in air. [4]
- 4 (a) A boat normally travels at 10 km/h in still water.
- Calculate the velocity of the boat observed from the shore if the boat moves
- (i) down a river,
- (ii) up a river, 
- (iii) across a river,
- flowing at 10 km/h . [6]
- (b) A jumbo jet cruises at a constant velocity of $1\,000 \text{ km/h}$ when the thrusting force of its engine is constant at 100 kN .
- State with a reason the
- (i) acceleration of the jet,
- (ii) force of air resistance on the jet. [4]

Q10:JUN 2013 P3

3 (a) Define the term

(i) *inertia*,

(ii) *velocity*.

[2]

(b) (i) A ball was projected at an angle of 30° above the horizontal with an initial velocity, V , in a vacuum. Another identical ball was projected at 60° to the horizontal from the same point in a vacuum.

1. Show that both balls had the same range.

2. Sketch, on the same axes, the path followed by each ball.

(ii) A boy at the top of a tower, 4.905 m high, threw a ball horizontally as shown in Fig. 3.1

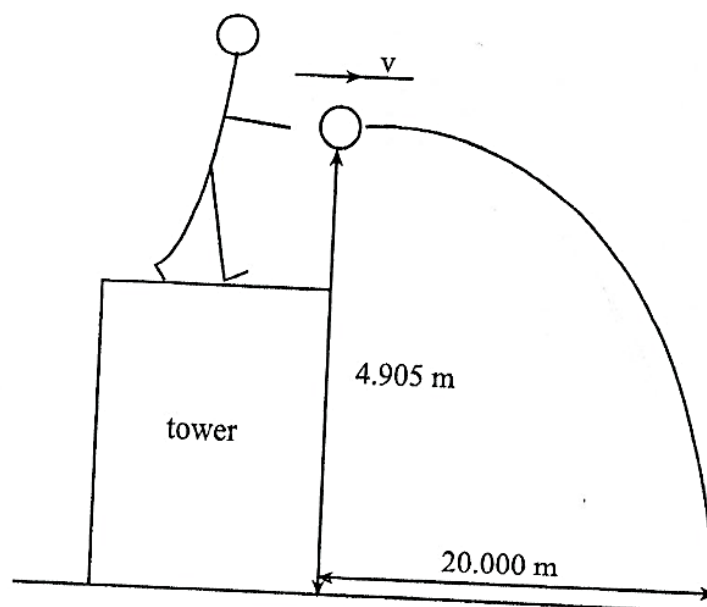


Fig. 3.1

Calculate the velocity, v , of the ball as it left the hand of the boy.

[6]

(c) A ball was launched in the earth's atmosphere at an angle θ to the horizontal.

Describe, with the aid of a diagram or otherwise, the effects of the atmosphere on the range and maximum height of projection.

[2]

Q11: NOV 2014 P3

- 3 (b) A stone thrown horizontally from the top of a cliff with a velocity of 15 ms^{-1} is observed to strike the ground at a distance 45 m from the base of the cliff.

Determine

- (i) the height of the cliff,
 - (ii) the velocity of the stone at the moment of impact. [6]
- (c) Explain why the path of the stone in (b) is not a straight line. [1]

Q12: NOV 2015 P3

- 2 (a) Distinguish between a vector quantity and a scalar quantity. [2]

- (b) A car travelling at 20 ms^{-1} due East turns and continues at 15 ms^{-1} due South.

- (i) Draw a vector diagram to illustrate the

- 1. initial and the final velocity,
- 2. change in velocity.

- (ii) Calculate the change in the

- 1. speed of the car,
- 2. velocity of the car.

[6]

9188/3 NOV15

- (c) Explain why the acceleration of an object may not be in the same direction as the velocity of the object. [2]

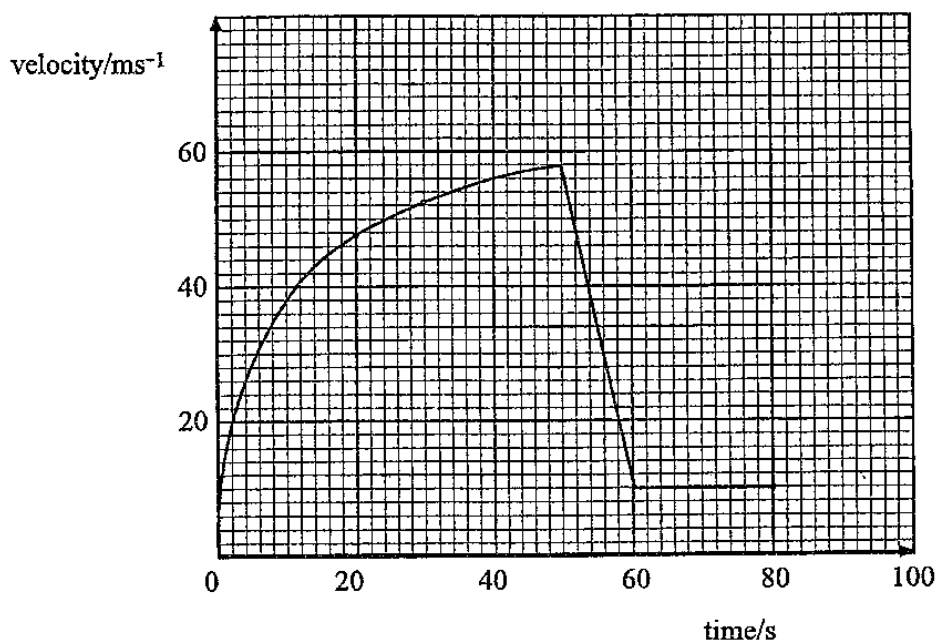
2.1 KINEMATICS P2&P3 QUESTIONS

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Q13:JUN 2017 P3

- 2 (a) Define *acceleration*. [1]

- (b) Fig. 2.1 shows a velocity-time graph for a parachutist after jumping from an aircraft.



- (i) State what is happening to the acceleration of the parachutist during the first 35 seconds.
- (ii) Explain your description in (i).
- (iii) Calculate the distance travelled between 60 seconds to 80 seconds. [6]
- (c) The distance from Harare to Beitbridge is 5.80×10^5 m by road. It takes 8 hours for a vehicle to travel this distance.
- (i) Define *distance*.
- (ii) Calculate the average speed of the vehicle.
- (iii) State **one** reason why practically it is impossible to maintain a constant velocity throughout the journey. [3]

Q14: JUN 2018 P3

- 2 Fig.2.1 shows how the velocity of an object dropped from a hovering helicopter at high altitude varies with time.

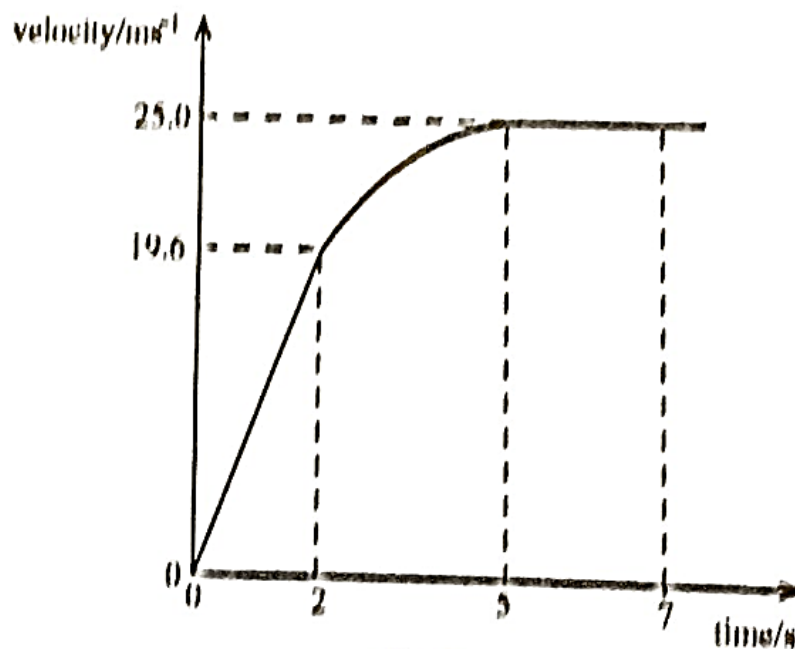


Fig.2.1

- (a) State how distance moved can be obtained from a velocity-time graph. [1]
- (b) (i) Calculate the
1. change in height,
 2. acceleration in the first 2 seconds.
- (ii) Explain the motion of the object from
1. 0s to 2s,
 2. 2s to 5s,
 3. 5s to 7s.
- (c) Explain how a parachute would ensure a safer landing of the object, [6]
- [3]

Q15: NOV 2018 P3

- 1 (c) (i) Derive, using the definition of velocity and acceleration the equation:

$$v^2 = u^2 + 2as$$

- (ii) A motorist travelling at 34 ms^{-1} spots a 30 ms^{-1} speed limit sign 50 m ahead. The reaction time for the driver is 0.2 s . On application of the brakes, the motorist decelerates uniformly at 5.0 ms^{-2} .

Deduce whether the motorist will attain the speed limit before reaching the sign.

[9]

Q16: 2018 SPEC P3

- (b) Fig. 1.2 shows the paths of a ball thrown against a vertical wall.

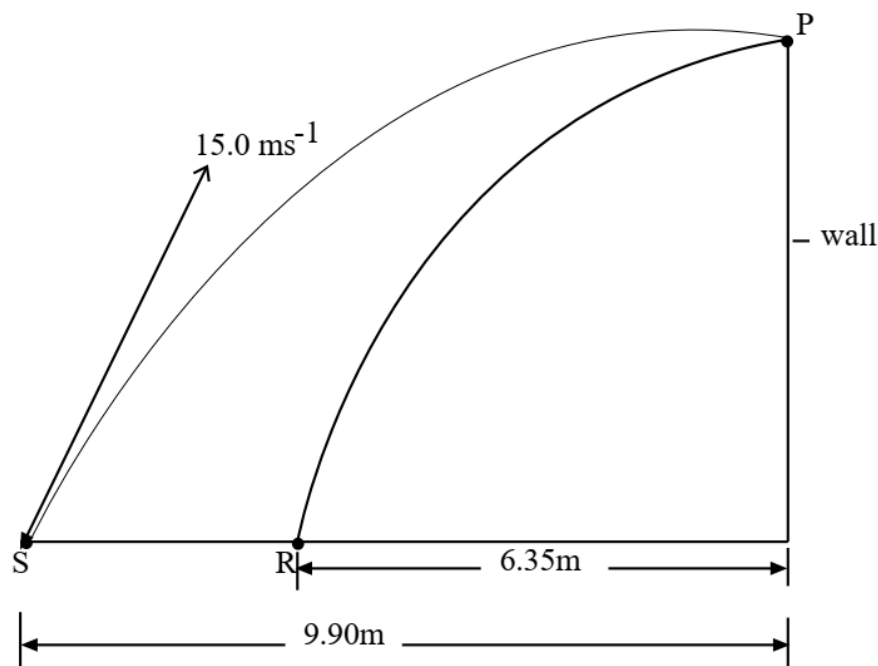


Fig. 1.2

The ball is thrown from S with an initial velocity of 15.0 ms^{-1} at 60° to the horizontal. The horizontal distance from S to the wall is 9.90 m . The ball hits the wall at P with a velocity at right angles to the wall and rebounds to point R.

Assume no air resistance.

- (i) Determine the vertical height gained by the ball as it travels from S to P.

2.1 KINEMATICS P2&P3 QUESTIONS

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- (ii) Calculate the time taken by the ball to travel from S to P.
- (iii) Show that the velocity of the ball immediately after rebounding from the wall is about 4.80 ms^{-1} .

Q17:NOV 2017 P3

- (b) A projectile of mass 250 kg is projected horizontally with a velocity of 1.5 ms^{-1} from a height of 5.0 m above the ground.

- (i) Define *work*.
- (ii) Calculate the
 1. speed of the projectile just before striking the ground,
 2. energy of the projectile just before striking the ground.
(Assume air resistance is negligible)
- (iii) State whether or not the body was in accelerated motion.
Explain your answer.

[6]